

MASALALAR YECHISH

13-

mavzu

1 Velosipedchi birinchi 4 s davomida 40 m, so'ng 8 s ichida 110 m va nihoyat yana 6 s ichida 30 m yo'l bosib o'tdi. Velosipedchining o'rtacha tezligini aniqlang.

Berilgan:	Formula	Hisoblash
$t_1 = 4 \text{ s}$ $s_1 = 40 \text{ m}$ $t_2 = 8 \text{ s}$ $s_2 = 110 \text{ m}$ $t_3 = 6 \text{ s}$ $s_3 = 30 \text{ m}$	$s = s_1 + s_2 + s_3$ $t = t_1 + t_2 + t_3$ $v_{or} = \frac{s}{t}$	$s = 40 \text{ m} + 110 \text{ m} + 30 \text{ m} = 180 \text{ m}$ $t = 4 \text{ s} + 8 \text{ s} + 6 \text{ s} = 18 \text{ s}$ $v_{or} = \frac{180 \text{ m}}{18 \text{ s}} = 10 \frac{\text{m}}{\text{s}}$ Javob: $v_{or} = 10 \frac{\text{m}}{\text{s}}$
Topish kerak: $v_{or} = ?$	$[v_{or}] = \frac{\text{m}}{\text{s}}$	

2 Jism yo'lning birinchi yarmini $6 \frac{\text{m}}{\text{s}}$, ikkinchi yarmini ikki marta tezroq bosib o'tgan bo'lsa, jismning butun harakat davomidagi o'rtacha tezligini toping.

Berilgan:	Formula	Hisoblash
$v_1 = 6 \frac{\text{m}}{\text{s}}$ $v_2 = 2v_1 = 12 \frac{\text{m}}{\text{s}}$ $s_1 = s_2 = \frac{s}{2}$ bunda s – to'liq harakat yo'li	$v_1 = \frac{s_1}{t_1}$ $v_2 = \frac{s_2}{t_2}$ $t_1 = \frac{s_1}{v_1} = \frac{s}{2v_1}$ $t_2 = \frac{s_2}{v_2} = \frac{s}{2v_2}$ $v_{or} = \frac{s_1 + s_2}{t_1 + t_2} = \frac{s}{\frac{s}{2v_1} + \frac{s}{2v_2}} = \frac{2v_1 \cdot v_2}{v_1 + v_2}$ $[v_{or}] = \frac{\text{m}}{\text{s}}$	$v_{or} = \frac{2 \cdot 6 \cdot 12}{6 + 12} \cdot \frac{\text{m}}{\text{s}} = 8 \frac{\text{m}}{\text{s}}$ Javob: $v_{or} = 8 \frac{\text{m}}{\text{s}}$
Topish kerak: $v_{or} = ?$		

3 Jism harakat vaqtining birinchi yarmida $15 \frac{\text{m}}{\text{s}}$, ikkinchi yarmida esa $10 \frac{\text{m}}{\text{s}}$ tezlik bilan harakatlangan. Jismning o'rtacha tezligini aniqlang.

Berilgan:	Formula	Hisoblash
$v_1 = 15 \frac{\text{m}}{\text{s}}$ $v_2 = 10 \frac{\text{m}}{\text{s}}$ $t_1 = t_2 = \frac{t}{2}$ bunda t – to'liq harakat vaqti Topish kerak: $v_{o'r} = ?$	$s_1 = v_1 \cdot t_1 \quad \text{va} \quad s_2 = v_2 \cdot t_2$ $v_{o'r} = \frac{s_{um}}{t_{um}} = \frac{s_1 + s_2}{t_1 + t_2}$ $v_{o'r} = \frac{v_1 \cdot t_1 + v_2 \cdot t_2}{t_1 + t_2} = \frac{v_1 \cdot t_1 + v_2 \cdot t_1}{t_1 + t_1} = \frac{v_1 + v_2}{2}$ $[v_{o'r}] = \frac{\text{m}}{\text{s}}$	$v_{o'r} = \frac{15 + 10}{2} \cdot \frac{\text{m}}{\text{s}} = 12,5 \frac{\text{m}}{\text{s}}$ Javob: $v_{o'r} = 12,5 \frac{\text{m}}{\text{s}}$



7-mashq



1 Avtomobil 4 soat davomida harakatlandi. Uning shu yo'ldagi o'rtacha tezligi $50 \frac{\text{km}}{\text{h}}$ ga teng. Avtomobil butun harakat davomida yurgan yo'lni aniqlang.

2 Jism 15 s davomida $5 \frac{\text{m}}{\text{s}}$, 10 s davomida $8 \frac{\text{m}}{\text{s}}$ va 5 s davomida $20 \frac{\text{m}}{\text{s}}$ tezlik bilan harakatlandi. Jismning o'rtacha harakat tezligi qanday bo'lgan?

3 Avtomobil ombordan yuk olib keldi. U omborga borishda $90 \frac{\text{km}}{\text{h}}$, qaytishda esa $60 \frac{\text{km}}{\text{h}}$ tezlik bilan harakatlandi. Avtomobilning harakat davomidagi o'rtacha tezligini toping.

4 Poyezd yo'lining birinchi yarmini ikkinchi yarmiga nisbatan 8 marta katta tezlikda bosib o'tdi. Agar poyezdning butun yo'l davomidagi o'rtacha tezligi $32 \frac{\text{km}}{\text{h}}$ bo'lsa, yo'lining ikkinchi yarmidagi tezligi qanday bo'lgan?

5 Avtomobil dovonga ko'tarilishda $10 \frac{\text{m}}{\text{s}}$, tushishda $15 \frac{\text{m}}{\text{s}}$ tezlik bilan harakatlandi. Tushish yo'li chiqish yo'lga teng bo'lsa, avtomobilning butun yo'l davomidagi o'rtacha tezligi qanday bo'lgan?